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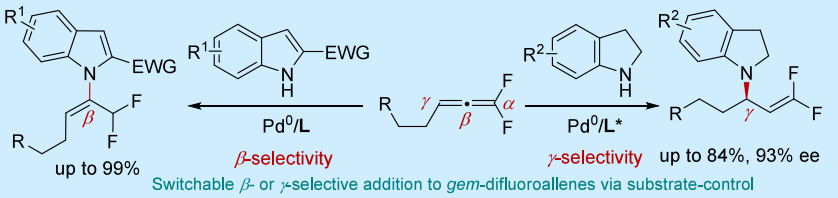
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Palladium-Catalyzed Substrate-Controlled Regiodivergent Hydroamination of *gem*-Difluoroallenes

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Switchable *β*- or *γ*-selective addition to *gem*-difluoroallenes via substrate-control

ABSTRACT: We report palladium-catalyzed regiodivergent hydroamination reactions of *gem*-difluoroallenes. Employing indolines as the nucleophiles leads to the formation of enantioenriched *γ*-addition products, while the use of 2-indole-carboxylate-type substrates switches the selectivity, delivering *β*-addition *E*-alkenes exclusively. This protocol provides efficient access to diverse difluoroalkylated indole derivatives, showing substantial substrate scope and broad functional group compatibility.

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gem-Difluoroalkenes and alkanes have gained significant attention in pharmaceutical chemistry owing to their enhanced metabolic properties and improved lipophilicity.[1] Among these, *N*-difluoropropylindoles and *N*-difluoroethylindoles have demonstrated considerable potential for broad bioactivities, including inhibitors of ASH, MAGL, and PAD, as well as serving as p53 isosteres (Scheme 1a).[2] As a result, the incorporation of difluoroalkyl groups into the indole frameworks has received growing interest, as it provides valuable opportunities for the development of novel therapeutic agents.

Traditional methods for the synthesis of *gem*-difluoroalkylated indoles typically employ nucleophilic substitution reactions between NH-free indoles and halogenated difluoroalkenes under basic conditions (Scheme 1b).[3] However, this approach is not atom-economical, as it results in the loss of a halogen atom and the use of excess bases. In contrast, transition metal-catalyzed hydroamination reaction of 1,1-difluoroallenes offers an efficient strategy for the introduction of difluoroallylic groups.[4] Although the reaction is promising, the asymmetric variants remain limited.[5] Notably, Shi and Wang developed a rhodium-catalyzed asymmetric hydroamination reaction of 1,1-difluoroallenes via exclusive *γ*-regioselective addition, whereas only one enantioenriched difluoroallylated indoline was furnished (Scheme 1c).[6]

With our continuous interest in Pd-catalyzed hydrofunctionalization of unsaturated hydrocarbons,[7] here we would like to present regiodivergent hydroamination reactions of 1,1-difluoroallenes by tuning the nucleophiles. This method enabled the asymmetric synthesis of *γ*-regioselective adducts when indolines were used as the nucleophiles, while *β*-regioselective products were obtained with the 2-indole-carboxylate-type substrates. The current strategy offered a versatile and efficient approach for constructing diverse difluoroalkylated indole derivatives, facilitating the rapid synthesis of structurally varied bioactive analogues.

The initial attempt was conducted by using 1,1-difluoroallene **1a** and indoline **2a** as the model substrates, under the catalysis of $Pd_2(dba)_3$ and **L1** [(*S*)-Segphos)] in toluene at 80 °C (Table 1, entry 1). While product **3a** was generated with exclusive *γ*-regioselectivity, the reaction suffered from low yield and enantioselectivity. It was found that product **3a** gradually decomposed during the reaction, giving unidentified side products. Then a variety of chiral ligands were tested, but phosphine-based ones generally led to poor yields.[8] In contrast, using bis(oxazoline) (BOX) ligand **L2** provided significantly improved enantiocontrol (entry 2). Notably, higher reactivity was observed with indane-derived ligands (**L3**−**5**) (entries 3−5), and a good ee value was obtained with **L5** having bulkier substituents (entry 5). Subsequently, some solvents were screened (entries 6−9), and DCM turned out to be the optimal one (entry 6). The reaction still proceeded well at a lower temperature, and slightly increased enantioselectivity was gained at 0 °C (entries 10 and 11). Additionally, a lower concentration was found to be beneficial for the yield, and the addition of 4 Å molecular sieves (MS) further improved the results (entries 12 and 13). Finally, other Pd sources, such as $Pd(PPh_3)_4$ and $Pd(OAc)_2$, proved to be ineffective (entries 14 and 15).

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Scheme 1. Biologically Active *gem*-Difluoroalkylated Indoles and Related Synthetic Strategies

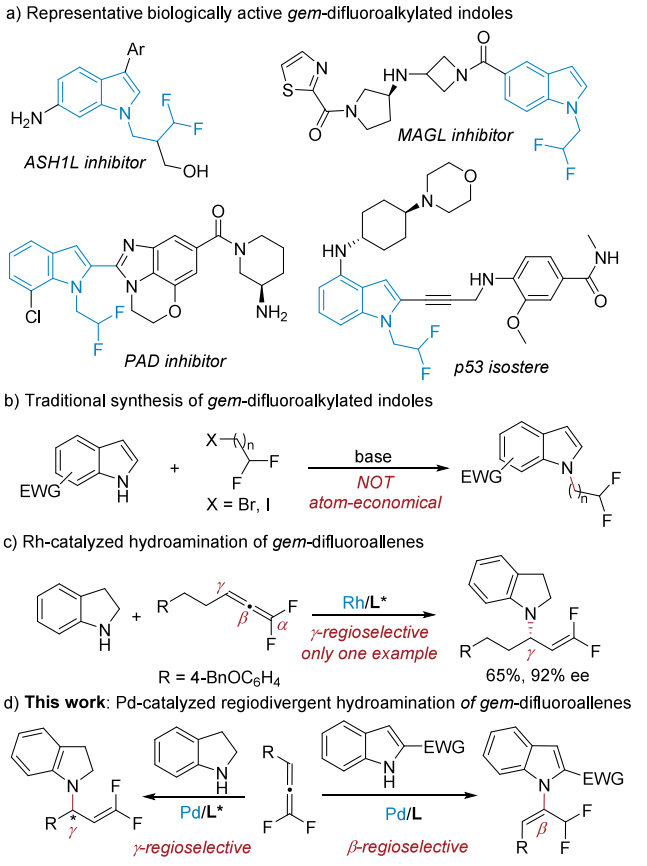
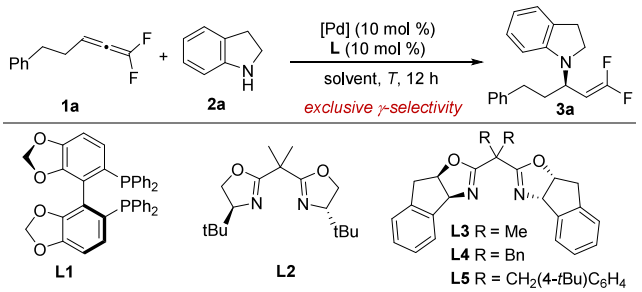


Table 1. Condition Optimizations for the γ-Regioselective Asymmetric Hydroamination[a]



entry	[Pd]	L	solvent	T (°C)	yield (%)[b]	ee (%)[c]
1	$Pd_2(dba)_3$	**L1**	toluene	80	33	4
2	$Pd_2(dba)_3$	**L2**	toluene	80	30	50
3	$Pd_2(dba)_3$	**L3**	toluene	80	36	26
4	$Pd_2(dba)_3$	**L4**	toluene	80	58	77
5	$Pd_2(dba)_3$	**L5**	toluene	80	55	80
6	$Pd_2(dba)_3$	**L5**	DCM	80	53	83
7	$Pd_2(dba)_3$	**L5**	EtOAc	80	46	65
8	$Pd_2(dba)_3$	**L5**	THF	80	63	42
9	$Pd_2(dba)_3$	**L5**	EtOH	80	messy	/
10	$Pd_2(dba)_3$	**L5**	DCM	40	50	82
11	$Pd_2(dba)_3$	**L5**	DCM	0	50	88
12[d]	$Pd_2(dba)_3$	**L5**	DCM	0	65	88
13[d,e]	$Pd_2(dba)_3$	**L5**	DCM	0	70	91
14	$Pd(PPh_3)_4$	**L5**	DCM	0	<5	/
15	$Pd(OAc)_2$	**L5**	DCM	0	<5	/

[a]Unless noted otherwise, the reaction was performed with **1a** (0.2 mmol), **2a** (0.1 mmol), $Pd_2(dba)_3$ (5 mol %) and **L** (10 mol %) in solvent (1.0 mL) at 80 °C for 12 h under Ar. [b]Yield of the isolated product. [c]Determined by HPLC analysis on a chiral stationary phase. [d]In DCM (2.0 mL). [e]With 4 Å MS (10 mg).

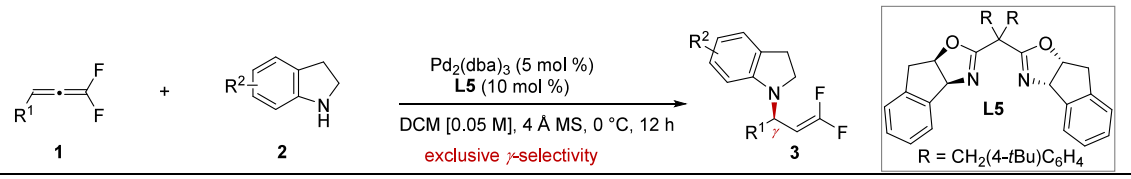
With the optimized conditions in hand, we then evaluated the substrate scope of the γ-regioselective asymmetric hydroamination of *gem*-difluoroallenes **1** and indolines **2** under the catalysis of $Pd_2(dba)_3$ and ligand **L5**. As shown in Scheme 2, excellent γ-regioselectivity was generally observed. Initial examples with diverse indoline derivatives revealed broad functional group tolerance. Both electron-donating and electron-withdrawing substituents on the aryl ring, including halogen atoms, were well accommodated, delivering γ-selective adducts **3b−g** with good to excellent enantioselectivity.[9] Notably, excellent enantioselectivity was maintained on a 1.0 mmol scale (product **3a**). Nevertheless, simple anilines and aliphatic secondary amines did not yield the desired products under the current reaction conditions.[8] Next, we explored various *gem*-difluoroallenes bearing diverse aryl substituents in the reactions with indoline **2a**. Substrates with a variety of substituents underwent the asymmetric transformation smoothly, furnishing enantioenriched products **3h−w** in moderate to good yields with uniformly high enantioselectivity. Importantly, a range of functional groups, including protected phenols, acetals, esters, and ketones, were well tolerated. Furthermore, simple alkyl-substituted allenes proved compatible, affording **3x−z** in moderate yields with good enantioselectivity. In addition, more complex substrates derived from citronellal and diosgenin were employed, and corresponding products **3aa** and **3ab** were furnished in good yields and stereoselectivity.

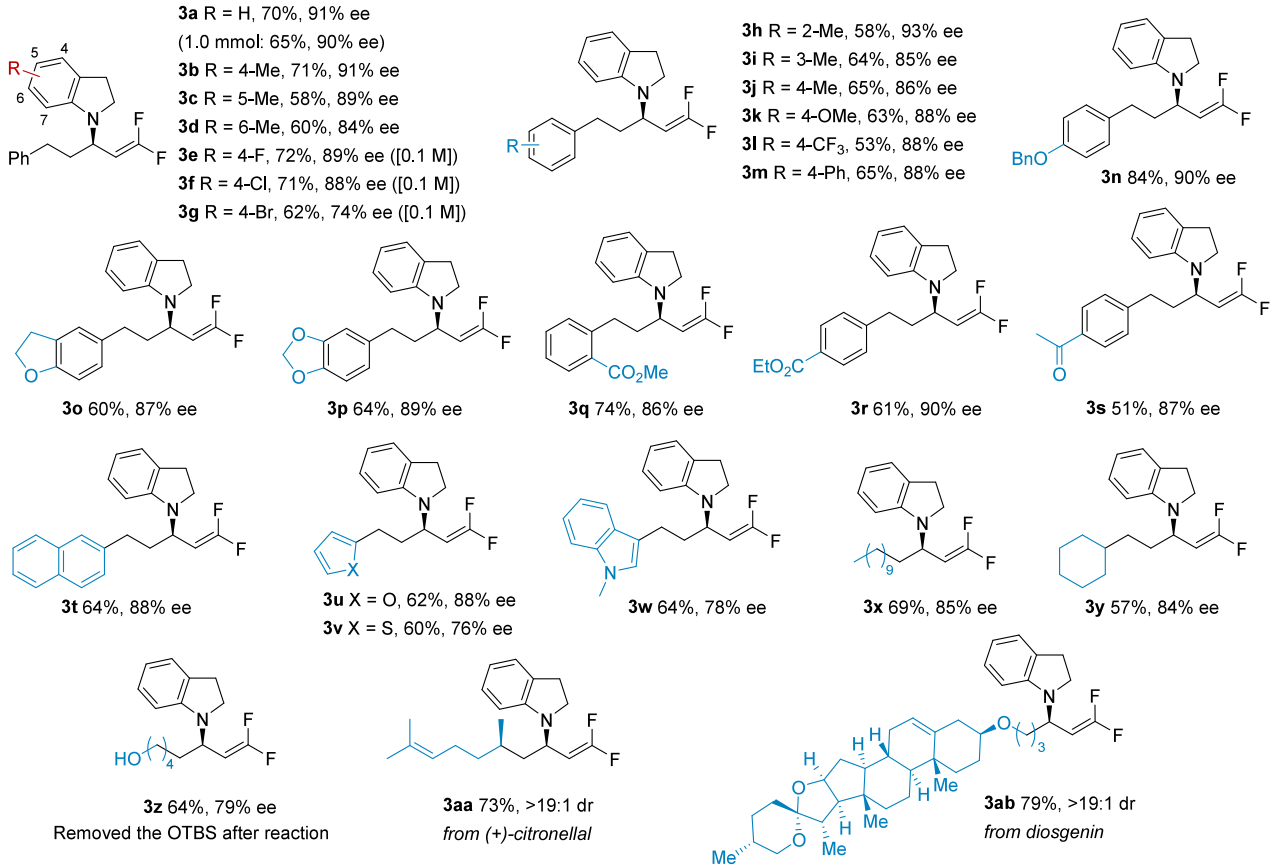
Interestingly, when 2-indole-carboxylates **4** were employed in the reactions with *gem*-difluoroallene **1a**, the regioselectivity was completely switched under the catalysis of $Pd_2(dba)_3$ and *rac*-**L1**,[8] delivering β-selective hydroamination product **5a−c** in moderate yields with exclusive *E*-selectivity (Scheme 3). Subsequent exploration revealed that a variety of substrates having relatively strong acidic NH groups well participated in similar hydroamination reaction, furnishing adducts **5d−h** in good to excellent yields. Importantly, a tetrasubstituted allene was applied to give **5i** in a high yield efficiently. Moreover, the reaction was compatible for diverse *gem*-difluoroallenes **1**, consistently affording products **5j−p** with good efficiency.

To exhibite the synthetic utility of this methodology, several transformations with the hydroamination products were conducted (Scheme 4). Using *gem*-difluoroallene **1a** as the precursor, γ-selective adduct **3a** from indoline and β-selective product **5d** from 2-indole-carboxylate **4a** could be produced under Pd catalysis, and valuable *N*-difluoropropylindole **6** and *N*-difluoroethylindole **7**, respectively, were accessed via simple transformations, showcasing the power of current protocol to construct structurally divergent difluoroalkylated indoles from the same starting material. Furthermore, adduct **3a** served as a versatile synthon for furnishing diverse fluorine-containing scaffolds. α-Fluorovinyl thioether **8** was obtained in a high yield with exclusive *E*-selectivity by simple treatment with ethyl thiol,[10] and trifluoromethyl product **9** was efficiently yielded via addition of TBAF.[11] In addition, controllable hydrodefluorination using either $LiAlH_4$ or Cu/bis(pinacolato)diboron (B_2pin_2) selectively provided *E*- and *Z*-monofluoroalkenes **10** and **11**, respectively, with retained ee values.[12] These transformations established complementary routes to

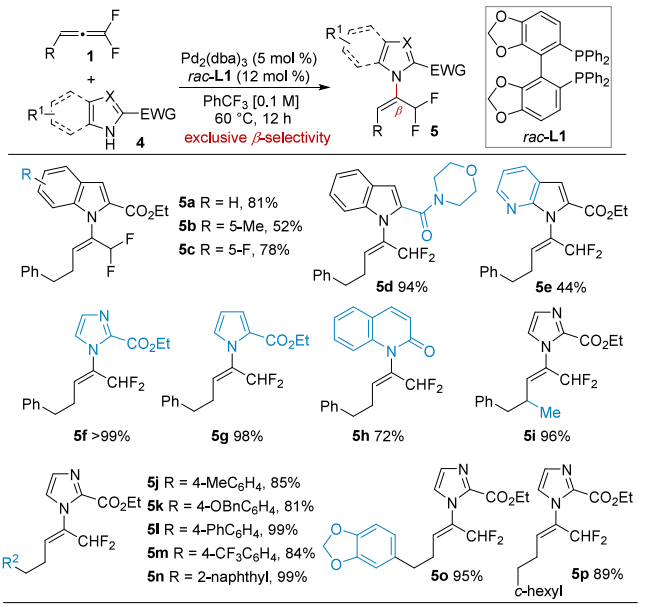
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Scheme 2. Substrate Scope for the Asymmetric γ-Selective Addition to *gem*-Difluoroallenes with Indolines

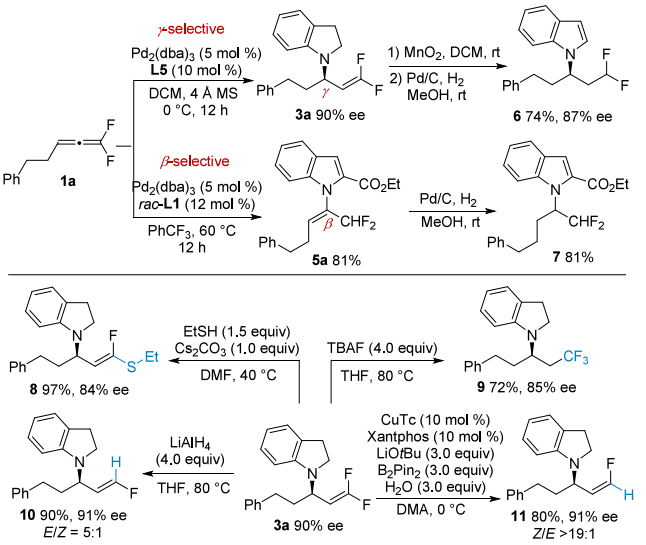




Scheme 3. Substrate Scope for the β-Selective Addition



Scheme 4. Transformations of Products

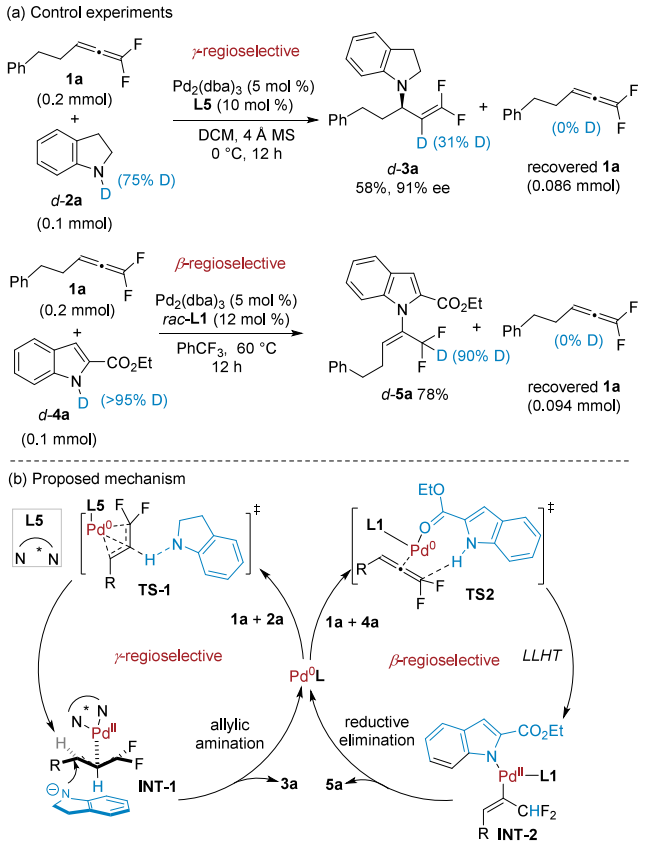


various fluorinated alkenes from a single precursor, significantly expanding the synthetic utility of our methodology.

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To gain more insight into the Pd-catalyzed reactions, some control experiments were conducted. In the case of γ-regioselective amination, Pd(II) proved ineffective (Table 1, entry 15), suggesting the reaction is initiated by Pd(0) species. When deuterium-labeled *d*-**2a** was applied to the asymmetric hydroamination reaction with $Pd_2(dba)_3$/**L5**, *d*-**3a** was afforded with 31% *D*-incorporation, along with recovered undeuterated **1a** (Scheme 5a). Given that indoline **2a** is a weak

Scheme 5. Mechanism Studies and the Proposed Mechanism of Hydroamination Reaction



acid, the related H−Pd species (via oxidative addition) is unlikely involved in this transformation.[13] As a result, it is proposed that *gem*-difluoroallene **1a** first coordinates to Pd(0), followed by protonation[14] with **2a** via transition state **TS-1** to generate π-allylpalladium intermediate **INT-1**, which then is attacked by indoline anion from the bottom face to afford product **3a** (Scheme 5b).[15] In contrast, the β-regioselective amination exhibited different behavior. When **1a** reacted with *d*-**4a** under the catalysis of $Pd_2(dba)_3$/*rac*-**L1**, high *D*-incorporation at α-site was observed. The results align with a ligand-to-ligand hydrogen transfer (LLHT) process[16] via **TS-2**, forming σ-vinyl-Pd intermediate **INT-2**, which subsequently undergoes reductive elimination to yield product **5a**.

In conclusion, we have investigated Pd-catalyzed regiodivergent hydroamination reactions of *gem*-difluoroallenes that demonstrated high nucleophile-dependent regioselectivity, efficiently constructing two distinct classes of valuable difluoroalkylated indole derivatives. By employing indolines as the nucleophiles, enantioenriched γ-addition products were obtained through a proposed Pd-mediated protonation pathway followed by asymmetric allylic amination, while 2-indole-carboxylate-type nucleophiles favored β-addition products likely via a LLHT process. This regiodivergent protocol provided a versatile synthetic platform for the rapid construction of structurally diverse difluoroalkylated indole derivatives, which might be applied as pharmaceutically relevant scaffolds in drug discovery.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its online Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.5c02471.

Complete experimental procedures and characterization of new products, synthetic transformations of the products, NMR and HRMS spectra, and HPLC chromatograms (PDF)

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Notes

The authors declare no competing financial interest.

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